

GARP Risk & AI — Q351–Q450

Unofficial Exam Prep

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Question

Which access control model uses attributes of the user, the resource, and the environment to make authorisation decisions?

- A. Role-Based Access Control (RBAC).
- B. Attribute-Based Access Control (ABAC).
- C. Mandatory Access Control (MAC).
- D. Discretionary Access Control (DAC).

Answer: B

Explanation

ABAC uses attributes of user, resource, and environment. RBAC uses roles; MAC uses security labels; DAC lets owners set permissions.

Question

Which of the following is NOT a data protection principle commonly found in privacy regulations?

- A. Data minimisation.
- B. Purpose limitation.
- C. Storage limitation.
- D. Unlimited retention.

Answer: D

Explanation

Modern privacy regulations require storage limitation; data must not be kept longer than necessary. Unlimited retention violates this principle.

Question

Which is the primary purpose of encrypting sensitive data both at rest and in transit?

- A. To increase storage capacity.
- B. To protect confidentiality and integrity against unauthorised access.
- C. To reduce processing time.
- D. To comply with marketing standards.

Answer: B

Explanation

Encryption protects data confidentiality and integrity. Options A, C, and D are unrelated.

Question

Which of the following is typically part of a model governance policy?

- A. The definition of a model and non-model.
- B. The model risk tiering methodology.
- C. Validation and documentation requirements.
- D. All of the above.

Answer: D

Explanation

Model governance policies define models, tiers, validation, documentation, monitoring, and reporting. All three options are standard components.

Question

Which item is essential in a model inventory?

- A. Only the model name.
- B. Model owner, purpose, risk tier, validation status, and interdependencies.
- C. The model's full source code.
- D. The model's marketing brochure.

Answer: B

Explanation

The inventory tracks ownership, purpose, risk tier, validation, and dependencies. Options A, C, and D are insufficient or inappropriate.

Question

What does a model landscape primarily provide that a model inventory does not?

- A. A list of all models.
- B. A map of interactions and interdependencies, highlighting cascade risk and critical nodes.
- C. Model code.
- D. Model marketing material.

Answer: B

Explanation

Landscape captures interactions; inventory lists models. Options A, C, and D describe inventory or unrelated content.

Question

Which of the following is a standard factor in AI/ML model risk tiering?

- A. Materiality of model use.
- B. Complexity of methodology.
- C. Exposure (financial, regulatory, reputational).
- D. All of the above.

Answer: D

Explanation

Risk tiering considers materiality, complexity, and exposure. All three are essential inputs.

Question

What is the primary purpose of independent model validation?

- A. To approve the model without further review.
- B. To critically and objectively assess the model's conceptual soundness, data quality, implementation, and performance, by parties independent of the developers.
- C. To rewrite the model code.
- D. To market the model.

Answer: B

Explanation

Independent validation provides objective challenge. Options A, C, and D are contrary or unrelated.

Question

Which of the following best describes when model validation should occur?

- A. Only at model retirement.
- B. Before deployment, periodically during use, and after material changes.
- C. Only after a serious incident.
- D. Once, at design time.

Answer: B

Explanation

Validation is continuous: initial, periodic, and change-based. Options A, C, and D are too narrow.

Question

The “use test” in model validation is best described as:

- A. A quantitative accuracy metric.
- B. A qualitative assessment of whether the model is appropriately used for its intended purpose in practice.
- C. A test of model speed.
- D. A code-review technique.

Answer: B

Explanation

The use test is qualitative and focused on real-world use. Options A, C, and D are different.

Question

Why must model use limits be documented and enforced?

- A. Because they are optional.
- B. Because using a model outside its validated domain can produce materially incorrect results and unexpected risk.
- C. Because they speed up training.
- D. Because they reduce storage.

Answer: B

Explanation

Use limits protect against out-of-scope use. Options A, C, and D are incorrect.

Question

A material change to a production model triggers:

- A. Immediate deployment without review.
- B. A change management process including impact assessment, documentation, validation, and approval.
- C. Retirement of the model.
- D. No action.

Answer: B

Explanation

Material changes go through structured change management. Options A, C, and D violate governance.

Question

Ongoing model monitoring should detect:

- A. Only training errors.
- B. Performance degradation, data drift, concept drift, and other changes that could affect model reliability.
- C. Only model speed.
- D. Only storage issues.

Answer: B

Explanation

Monitoring detects degradation and drift. Options A, C, and D are too narrow.

Question

Which statement correctly distinguishes data drift from concept drift?

- A. They are the same.
- B. Data drift = change in input distribution; concept drift = change in relationship between inputs and target.
- C. Data drift = change in labels only.
- D. Concept drift = change in storage.

Answer: B

Explanation

Data drift: input distribution shift. Concept drift: feature-target relationship shift. Options A, C, and D misstate.

Question

Which of the following is the **FIRST** step in the model development process?

- A. Deploy the model.
- B. Define objectives and scope.
- C. Retire the model.
- D. Audit the model.

Answer: B

Explanation

Development starts with clear objectives and scope. Options A, C, and D are subsequent or unrelated stages.

Question

During data collection and preprocessing, analysts should:

- A. Skip documentation.
- B. Identify sources, clean data, handle missing values, and document any exclusions with rationale.
- C. Ignore outliers.
- D. Use all data without inspection.

Answer: B

Explanation

Data preparation requires cleaning, missing-value handling, and documentation. Options A, C, and D skip essential steps.

Question

Feature engineering typically includes:

- A. Combining variables, deriving new features, transformations, and dimensionality reduction.
- B. Only removing features.
- C. Only scaling.
- D. Only imputation.

Answer: A

Explanation

Feature engineering is broad: transformations, new features, combinations, reductions. Options B, C, and D are partial.

Question

Model selection should be based on:

- A. Personal preference.
- B. A comprehensive evaluation of alternatives, considering performance, interpretability, complexity, and business context.
- C. Speed alone.
- D. Cost alone.

Answer: B

Explanation

Model selection balances multiple criteria. Options A, C, and D are oversimplified.

Question

During model training, cross-validation is used to:

- A. Increase training time.
- B. Estimate generalisation performance and tune hyperparameters.
- C. Skip validation.
- D. Remove features.

Answer: B

Explanation

Cross-validation estimates generalisation and supports hyperparameter tuning. Options A, C, and D are incorrect.

Question

Developer testing before deployment is designed to:

- A. Skip validation.
- B. Evaluate performance before production release; ensure accuracy, robustness, and stability.
- C. Replace independent validation.
- D. Reduce code size.

Answer: B

Explanation

Developer testing precedes and complements independent validation. Options A, C, and D misstate its purpose.

Question

Which of the following is NOT a standard testing stage?

- A. Unit tests.
- B. Integration tests.
- C. Performance tests.
- D. Marketing tests.

Answer: D

Explanation

Standard testing: unit, component, integration, system, performance, regression, acceptance, adversarial. Marketing is not a testing stage.

Question

What is adversarial testing in the context of AI models?

- A. Standard unit testing.
- B. Actively probing the model with crafted inputs to discover unintended behaviour or vulnerabilities (red teaming).
- C. Testing hardware.
- D. Testing marketing materials.

Answer: B

Explanation

Adversarial testing = red teaming with crafted inputs. Options A, C, and D are unrelated.

Question

Regression testing ensures:

- A. New functionality works as expected.
- B. The model continues to function correctly after changes elsewhere in the system.
- C. Faster performance.
- D. Lower cost.

Answer: B

Explanation

Regression tests verify existing functionality after modifications. Options A, C, and D describe different goals.

Question

Acceptance testing determines whether:

- A. The model meets agreed-upon requirements and is ready for deployment.
- B. The model is the fastest available.
- C. The model is cheapest.
- D. The code is clean.

Answer: A

Explanation

Acceptance testing verifies readiness for deployment. Options B, C, and D are not the criterion.

Question

Which is a primary benefit of an AI ethics framework?

- A. Increased training speed.
- B. Clear guidance for evaluating ethical dilemmas, enhanced trust, and reduced regulatory non-compliance risk.
- C. Reduced model size.
- D. Eliminated bias.

Answer: B

Explanation

Ethics frameworks provide guidance and reduce regulatory risk. Options A, C, and D are incorrect or overstated.

Question

Which of the following best captures the concept of Responsible AI?

- A. Maximising model accuracy without regard for consequences.
- B. Ethical, transparent, accountable AI aligned with societal values and mitigating potential harms.
- C. Deploying AI as fast as possible.
- D. Ignoring regulations.

Answer: B

Explanation

Responsible AI encompasses ethics, transparency, accountability, and harm mitigation. Options A, C, and D contradict the concept.

Question

Which is a governance challenge for AI?

- A. Power asymmetries between large tech firms and regulators.
- B. Lack of interpretability.
- C. Unpredictability of AI systems.
- D. All of the above.

Answer: D

Explanation

Power asymmetries, opacity, and unpredictability are all governance challenges.

Question

Compared to medical ethics, what is missing from AI ethics?

- A. Nothing.
- B. Mature professional structures: licensing, mandatory training, ethics committees, and enforceable codes.
- C. Only training.
- D. Only licensing.

Answer: B

Explanation

AI ethics lacks the mature professional structures of medical ethics. Options A, C, and D are too narrow.

Question

The EU AI Act regulates AI based on:

- A. A single uniform standard.
- B. A risk-tiered approach: unacceptable, high, limited, minimal.
- C. Self-certification only.
- D. No requirements.

Answer: B

Explanation

The AI Act uses a risk-tiered approach. Options A, C, and D misstate the structure.

Question

Which best describes China's AI regulation?

- A. No AI regulation.
- B. Targeted regulations including the Provisional Administrative Measures for Generative AI Services and PIPL.
- C. Only medical AI regulation.
- D. Complete industry self-regulation.

Answer: B

Explanation

China has adopted targeted AI regulations. Options A, C, and D are incorrect.

Question

Fairness within Responsible AI is BEST described as:

- A. Treating all individuals identically
- B. Preventing unjustified bias and discrimination
- C. Maximizing model accuracy
- D. Maximizing automation

Answer: B

Question

Which statement correctly distinguishes consequentialism, deontology, and virtue ethics?

- A. All three are identical.
- B. Consequentialism focuses on outcomes; deontology on duties; virtue ethics on character.
- C. All three focus only on outcomes.
- D. All three ignore outcomes.

Answer: B

Explanation

The three frameworks differ in focus. Options A, C, and D misstate them.

Question

Which statement about justice is correct?

- A. Different distributive justice concepts can conflict.
- B. Impartiality is not required for procedural justice.
- C. Autonomy is secondary to social justice.
- D. Meritocracy is not a distributive justice concept.

Answer: A

Explanation

Distributive justice concepts can conflict. Options B, C, and D are incorrect.

Question

Which statement about justice is correct?

- A. Different distributive justice concepts can conflict.
- B. Impartiality is not required for procedural justice.
- C. Autonomy is secondary to social justice.
- D. Meritocracy is not a distributive justice concept.

Answer: A

Explanation

Different theories of distributive justice may produce conflicting outcomes. Equality, need, merit, and utility can result in different distributions.

Question

What best distinguishes equity from equality?

- A. Equality gives identical outcomes while equity considers relevant differences.
- B. Equity requires identical treatment.
- C. Equality always produces fair outcomes.
- D. Equity ignores historical disadvantage.

Answer: A

Explanation

Equality focuses on equal treatment, while equity may require different treatment to achieve fair opportunities or outcomes.

Question

Which principle is most central to procedural justice?

- A. Transparency and impartial decision making
- B. Maximum economic efficiency
- C. Equal outcomes
- D. Maximization of shareholder value

Answer: A

Explanation

Procedural justice focuses on whether decisions are made through fair, consistent, transparent, and unbiased processes.

Question

Under a merit-based conception of distributive justice, benefits should primarily be allocated according to:

- A. Individual contribution and achievement
- B. Random assignment
- C. Equal outcomes
- D. Social status

Answer: A

Explanation

Meritocracy allocates rewards based on contribution, skill, effort, or achievement.

Question

A need-based theory of justice would most likely support:

- A. Allocating resources according to individual need
- B. Equal distribution regardless of circumstances
- C. Allocation based solely on performance
- D. Allocation by lottery

Answer: A

Explanation

Need-based approaches prioritize those with greater needs rather than equal or merit-based distributions.

Question

What is a common challenge in algorithmic fairness?

- A. Multiple fairness metrics may be impossible to satisfy simultaneously.
- B. Fairness eliminates all model risk.
- C. Fairness guarantees equal outcomes.
- D. Transparency removes all bias.

Answer: A

Explanation

Different mathematical definitions of fairness can conflict, requiring governance decisions and tradeoff analysis.

Question

Corrective justice is primarily concerned with:

- A. Remedying harms or wrongs
- B. Equal distribution of resources
- C. Economic growth
- D. Maximizing efficiency

Answer: A

Explanation

Corrective justice seeks to restore parties after harm, error, discrimination, or other wrongdoing.

Question

Which action best promotes justice in AI systems?

- A. Assessing disparate impacts across affected groups
- B. Maximizing accuracy regardless of consequences
- C. Eliminating human oversight
- D. Ignoring stakeholder concerns

Answer: A

Explanation

Justice-oriented AI governance evaluates how benefits, burdens, and errors are distributed among stakeholders.

Question

Which is the best mitigation for hallucination risk in a legal-advice GenAI application?

- A. Increase temperature.
- B. Ground the model with retrieval-augmented generation (RAG) using authoritative sources, and require human review for critical outputs.
- C. Use a smaller model.
- D. Disable the model.

Answer: B

Explanation

RAG grounds output; human review catches remaining errors. Options A, C, and D are inadequate or extreme.

Question

Which is an internal governance lever for GenAI?

- A. Assigning clear ownership and accountability across the GenAI lifecycle.
- B. Tracking external regulatory changes.
- C. Relying only on external standards.
- D. Outsourcing all governance.

Answer: A

Explanation

Internal levers include ownership, review processes, escalation, training. Options B and C are external; D is contrary to governance.

Question

Which is an external governance constraint for GenAI?

- A. Evolving regulatory requirements.
- B. Contractual terms with model providers.
- C. Emerging standards (e.g., NIST AI RMF, ISO/IEC 42001).
- D. All of the above.

Answer: D

Explanation

All three are external governance constraints on GenAI deployments.

Question

When registering a GenAI system in a model inventory, which additional aspects should be captured beyond traditional models?

- A. The underlying foundation model, prompt templates, and any retrieval/knowledge sources.
- B. Only the API provider.
- C. Only the temperature parameter.
- D. Only the compute costs.

Answer: A

Explanation

GenAI inventory requires capturing foundation model, prompts, and retrieval data. Options B, C, and D are partial.

Question

Which is a unique challenge when validating GenAI models?

- A. Same challenges as traditional models.
- B. Open-ended output, hallucinations, multi-modality, and difficulty establishing ground truth.
- C. Only speed.
- D. Only cost.

Answer: B

Explanation

GenAI validation faces open-ended output, hallucination, multi-modality, and ground-truth issues. Options A, C, and D underestimate the challenge.

Question

Why does the AI/ML paradigm shift governance emphasis toward data?

- A. Data is irrelevant to AI.
- B. AI/ML models often learn directly from data, making data quality, provenance, and bias central to model behaviour and risk.
- C. Models do not depend on data.
- D. Governance is unchanged.

Answer: B

Explanation

AI/ML models learn from data; data governance is therefore central. Options A, C, and D are incorrect.

Question

Which attribute is most relevant when classifying AI/ML models for ethical review?

- A. Model accuracy only.
- B. Data bias, explainability, and recalibration needs, among others.
- C. Model speed.
- D. Compute cost.

Answer: B

Explanation

Ethical review focuses on bias, explainability, and recalibration. Options A, C, and D are secondary.

Question

A model landscape captures interdependencies because:

- A. Models operate in isolation.
- B. Outputs of one model are often inputs to others, so failures can cascade across an organisation.
- C. Dependencies are irrelevant.
- D. Only audit cares about them.

Answer: B

Explanation

Interdependencies create cascading risk. Options A, C, and D are incorrect.

Question

Aggregate model risk can be influenced by:

- A. Only individual model quality.
- B. Model interdependencies, shared assumptions, and correlated methodology.
- C. Only data size.
- D. Only IT infrastructure.

Answer: B

Explanation

Aggregate risk arises from interdependencies and shared assumptions. Options A, C, and D miss this.

Question

Why can aggregating data across sources introduce model risk?

- A. Aggregation always improves quality.
- B. Inconsistent definitions, timeframes, or measurement scales can distort features and lead to biased models.
- C. Aggregation eliminates risk.
- D. Aggregation is always safe.

Answer: B

Explanation

Aggregation can introduce inconsistencies. Options A, C, and D are incorrect.

Question

Which is an example of a production feedback loop creating model risk?

- A. A model trained on past decisions then used to make new decisions, which become the next training data.
- B. A model without historical data.
- C. A model that is never deployed.
- D. A model that is retired.

Answer: A

Explanation

Feedback loops reinforce biases. Options B, C, and D are not feedback loops.

Question

Why is data and model lineage important for AI governance?

- A. To track data and model transformations end-to-end, enabling reproducibility, auditability, and root-cause analysis.
- B. To speed up training.
- C. To reduce costs.
- D. To remove features.

Answer: A

Explanation

Lineage supports reproducibility and audit. Options B, C, and D are secondary.

Question

Data stewards are responsible for:

- A. Only fixing typos.
- B. Managing data quality, access, definitions, and use within a governance framework.
- C. Building models.
- D. Marketing data.

Answer: B

Explanation

Data stewards manage quality, access, and definitions. Options A, C, and D are too narrow or unrelated.

Question

Data custodians are primarily responsible for:

- A. Business ownership of data.
- B. Technical storage, security, and access controls for data.
- C. Building models.
- D. Marketing data.

Answer: B

Explanation

Custodians handle technical storage and access. Stewards handle business data quality; owners provide governance. Options A, C, and D are not custody roles.

Question

Data governance reporting to the board typically includes:

- A. Only storage costs.
- B. Data quality metrics, incidents, compliance status, and key risks.
- C. Only IT staff lists.
- D. Only marketing spend.

Answer: B

Explanation

Board-level reporting covers quality, incidents, compliance, and risk. Options A, C, and D are irrelevant.

Question

Which of the following is a trigger for ad-hoc model revalidation?

- A. Only scheduled review.
- B. Significant changes in inputs, environment, or performance deviations.
- C. Only cost increases.
- D. Only headcount changes.

Answer: B

Explanation

Ad-hoc revalidation is triggered by material changes or performance issues. Options A, C, and D are not standard triggers.

Question

Recalibration of a model is appropriate when:

- A. The model's structure is wrong.
- B. The core structure remains valid but parameters need updating in light of new data.
- C. The model is being retired.
- D. The model has never been used.

Answer: B

Explanation

Recalibration updates parameters; structural issues require redevelopment. Options A, C, and D describe different actions.

Question

A model risk appetite statement typically includes:

- A. Detailed code.
- B. The organisation's tolerance for model risk, escalation triggers, and consequences.
- C. Only model inventory.
- D. Only validation reports.

Answer: B

Explanation

Appetite statements define tolerance and triggers. Options A, C, and D are unrelated.

Question

Which of the following is NOT typically included in a model risk taxonomy?

- A. Conceptual risk.
- B. Data risk.
- C. Implementation risk.
- D. Employee satisfaction risk.

Answer: D

Explanation

Model risk taxonomies focus on conceptual, data, and implementation risk (plus usage risk). Employee satisfaction is not standard.

Question

A model risk register typically tracks:

- A. Only model code.
- B. Identified risks, owners, mitigation actions, deadlines, and status.
- C. Only training data.
- D. Only IT systems.

Answer: B

Explanation

A risk register tracks identified risks, owners, actions, deadlines, and status. Options A, C, and D are unrelated.

Question

A model risk committee is typically responsible for:

- A. Approving models and overseeing model risk governance at the enterprise level.
- B. Entering data.
- C. Training models.
- D. Marketing models.

Answer: A

Explanation

The model risk committee provides enterprise oversight and approvals. Options B, C, and D are operational roles.

Question

Effective model risk training programs should include:

- A. Only technical content.
- B. Technical content plus governance, ethics, and regulatory expectations.
- C. Only ethics.
- D. Only regulation.

Answer: B

Explanation

Model risk training should be comprehensive: technical, governance, ethics, regulation.

Question

A strong model risk culture is characterised by:

- A. Silence about risks.
- B. Awareness, accountability, open communication, and continuous learning around model risk.
- C. Only senior leaders caring about risk.
- D. Ignoring model limitations.

Answer: B

Explanation

A strong culture values awareness, accountability, openness, and learning. Options A, C, and D contradict this.

Question

Model risk reporting to governance bodies typically occurs:

- A. Only when problems arise.
- B. On a scheduled basis (e.g., quarterly) and ad hoc for significant issues.
- C. Never.
- D. Only at year-end.

Answer: B

Explanation

Reporting is scheduled and ad hoc as needed. Options A, C, and D are insufficient.

Question

Which of the following is a lawful basis for processing personal data under GDPR?

- A. Consent.
- B. Contract performance.
- C. Legal obligation.
- D. All of the above.

Answer: D

Explanation

GDPR Art. 6 specifies lawful bases: consent, contract, legal obligation, vital interests, public task, legitimate interests. Options A, B, and C are each valid bases.

Question

Which is a data subject right under GDPR?

- A. Right to access.
- B. Right to rectification.
- C. Right to erasure.
- D. All of the above.

Answer: D

Explanation

GDPR grants multiple rights: access, rectification, erasure, restriction, portability, objection. Options A, B, and C are each among them.

Question

Privacy by design requires:

- A. Adding privacy features at the end.
- B. Embedding privacy protections into the design and architecture of systems from the outset.
- C. Ignoring privacy until a complaint.
- D. Only privacy policies.

Answer: B

Explanation

Privacy by design embeds protections from the start. Options A, C, and D are inconsistent with the principle.

Question

What is the difference between anonymisation and pseudonymisation?

- A. They are the same.
- B. Anonymisation irreversibly removes identifiability; pseudonymisation replaces identifiers with a key that allows re-identification.
- C. Pseudonymisation is irreversible.
- D. Anonymisation always fails.

Answer: B

Explanation

Anonymisation is irreversible; pseudonymisation can be reversed with the key. Options A, C, and D misstate the distinction.

Question

What is the purpose of differential privacy?

- A. To remove all data.
- B. To allow statistical analysis of a dataset while protecting individual records, by adding calibrated noise.
- C. To encrypt the data.
- D. To compress data.

Answer: B

Explanation

Differential privacy adds noise to protect individual records while preserving aggregate statistics. Options A, C, and D are different techniques.

Question

Which of the following is typically part of an ethical AI review?

- A. Only model accuracy.
- B. Assessing fairness, transparency, accountability, and potential harms.
- C. Only speed.
- D. Only cost.

Answer: B

Explanation

Ethical AI reviews assess fairness, transparency, accountability, and harms. Options A, C, and D are insufficient.

Question

An algorithmic impact assessment (AIA) typically evaluates:

- A. Only model accuracy.
- B. Risks and impacts on individuals, groups, and society, especially for high-risk use cases.
- C. Only cost.
- D. Only speed.

Answer: B

Explanation

AIAs evaluate broader risks and impacts. Options A, C, and D are too narrow.

Question

Which is a limitation of impurity-based feature importance for tree ensembles?

- A. It is always accurate.
- B. It can be biased toward high-cardinality or correlated features.
- C. It cannot be computed.
- D. It replaces SHAP.

Answer: B

Explanation

Impurity-based importance is biased; permutation importance or SHAP are alternatives. Options A, C, and D are incorrect.

Question

Partial dependence plots show:

- A. Raw data.
- B. The marginal effect of a feature on the model's prediction, averaged over other features.
- C. Model code.
- D. Only training errors.

Answer: B

Explanation

PDPs display marginal feature effects. Options A, C, and D are unrelated.

Question

Which property does SHAP have that many other feature attribution methods lack?

- A. Non-determinism.
- B. Consistency — if a feature's contribution increases, its SHAP value does not decrease.
- C. Randomness.
- D. Irreproducibility.

Answer: B

Explanation

SHAP satisfies consistency, along with local accuracy and missingness. Options A, C, and D are not properties of SHAP.

Question

Which is a limitation of LIME?

- A. It is always globally faithful.
- B. It provides local approximations that may not be stable across similar instances and are not guaranteed to be globally faithful.
- C. It has no applications.
- D. It replaces model training.

Answer: B

Explanation

LIME is local and can be unstable. Options A, C, and D misstate.

Question

Counterfactual explanations are particularly useful because:

- A. They describe model code.
- B. They provide actionable, human-understandable explanations of what would need to change for a different decision.
- C. They replace SHAP.
- D. They measure accuracy.

Answer: B

Explanation

Counterfactuals provide actionable insights. Options A, C, and D are incorrect.

Question

When data drift is detected in a production model, which of the following is generally the appropriate response?

- A. Ignore it.
- B. Investigate cause, consider recalibration or retraining, and update governance documentation accordingly.
- C. Delete the model.
- D. Change the target.

Answer: B

Explanation

Drift triggers investigation and possible recalibration/retraining. Options A, C, and D are not standard.

Question

Which of the following best summarises the guiding principle of Responsible AI governance in financial services?

- A. Maximise model complexity.
- B. Balance performance with governance, validation, monitoring, fairness, explainability, and ethical alignment with regulatory and business requirements.
- C. Deploy all models without validation.
- D. Ignore risks.

Answer: B

Explanation

Responsible AI balances performance with governance, validation, monitoring, fairness, explainability, and ethics. Options A, C, and D contradict the principle.

Question

A bank wants to assess fairness immediately after making lending decisions. Loan defaults typically occur years later. Which fairness criterion is MOST practical at decision time?

- A. Predictive Parity
- B. Equal Opportunity
- C. Demographic Parity
- D. Equalized Odds

Answer: C

Explanation

Demographic parity does not require knowledge of future outcomes. The other metrics depend on ground-truth labels that become available only later.

Question

Why can demographic parity be problematic in medical diagnosis?

- A. It requires too much training data.
- B. It ignores legitimate differences in disease prevalence across groups.
- C. It requires causal models.
- D. It maximizes false negatives.

Answer: B

Explanation

If disease prevalence differs across groups, forcing equal positive prediction rates may reduce predictive usefulness and medical effectiveness.

Question

Which statement about demographic parity is TRUE?

- A. It guarantees equal error rates.
- B. It guarantees equal predictive value.
- C. It says nothing about whether predictions are correct.
- D. It guarantees equal opportunity.

Answer: C

Explanation

Demographic parity focuses only on equal prediction rates across groups and does not evaluate prediction quality.

Question

When base rates differ across groups, what does the fairness impossibility theorem imply?

- A. Demographic parity, predictive parity, and equal opportunity can generally be satisfied simultaneously.
- B. One fairness definition becomes objectively correct.
- C. Certain fairness criteria cannot all be satisfied at the same time.
- D. Fairness measurement becomes impossible.

Answer: C

Explanation

The theorem highlights that competing fairness definitions may conflict when underlying group base rates differ.

Question

A fraud model is deployed in an environment where confirmed fraud labels may take months to establish and many fraud cases are never confirmed. Why might demographic parity still be considered?

- A. It guarantees fairness.
- B. It does not require reliable ground-truth labels.
- C. It guarantees equal precision.
- D. It eliminates bias.

Answer: B

Explanation

Demographic parity can be evaluated without waiting for true fraud outcomes, making it attractive in settings where labels are delayed, incomplete, or extremely rare.

Question

Predictive rate parity requires equality of:

- A. Positive prediction rates
- B. Precision across groups
- C. Sensitivity across groups
- D. False positive rates across groups

Answer: B

Precision across groups

Explanation

Predictive rate parity (also called conditional use accuracy equality or predictive parity) requires that the positive predictive value (precision) be equal across groups:

$$P(Y = 1 \mid \hat{Y} = 1, A = 0) = P(Y = 1 \mid \hat{Y} = 1, A = 1)$$

In words: among all individuals who are *predicted* to be positive, the proportion who are *actually* positive must be the same for every group. This conditions on the prediction (the predicted positive class), not on the true label. It does *not* require equal positive prediction rates (that is demographic parity), nor equal sensitivity (that is equal opportunity), nor equal false positive rates (that is predictive equality).

Question

Equal opportunity requires equality of:

- A. Precision
- B. Overall accuracy
- C. Sensitivity (true positive rate)
- D. Positive prediction rate

Answer: C

Sensitivity (true positive rate)

Explanation

Equal opportunity requires that the true positive rate (sensitivity / recall) be equal across groups:

$$P(\hat{Y} = 1 \mid Y = 1, A = 0) = P(\hat{Y} = 1 \mid Y = 1, A = 1)$$

In words: among all individuals who are *truly* positive, the proportion who are *correctly* identified as positive must be the same for every group. This conditions on the true label (the actual positive class). It does *not* require equal precision (that is predictive rate parity), nor equal overall accuracy, nor equal positive prediction rates (that is demographic parity).

Question

Which fairness criterion conditions on the predicted positive class?

- A. Equal Opportunity
- B. Predictive Rate Parity
- C. Equalized Odds
- D. Demographic Parity

Answer: B

Predictive Rate Parity

Explanation

Predictive rate parity conditions on the predicted positive class ($\hat{Y} = 1$). It asks: given that the model predicted positive, what is the probability that the individual is truly positive?

$$P(Y = 1 \mid \hat{Y} = 1, A = 0) = P(Y = 1 \mid \hat{Y} = 1, A = 1)$$

This is exactly the definition of precision (positive predictive value). By contrast:

Equal Opportunity conditions on the *true* positive class ($Y = 1$) and equalizes sensitivity (TPR).

Equalized Odds conditions on *both* the true positive and true negative classes, equalizing TPR and FPR simultaneously.

Demographic Parity does not condition on any outcome label; it simply equalizes the positive prediction rate $P(\hat{Y} = 1 \mid A)$.

Question

Which fairness criterion conditions on those who are truly positive?

- A. Equal Opportunity
- B. Predictive Rate Parity
- C. Demographic Parity
- D. Predictive Equality

Answer: A

Equal Opportunity

Explanation

Equal opportunity conditions on the true positive class ($Y = 1$). It requires equal sensitivity (true positive rate) across groups:

$$P(\hat{Y} = 1 \mid Y = 1, A = 0) = P(\hat{Y} = 1 \mid Y = 1, A = 1)$$

In words: among those who are *truly* positive, the probability of being correctly predicted positive must be the same for all groups. Why the others are wrong:

Predictive Rate Parity conditions on the *predicted* positive class ($\hat{Y} = 1$), not the true positive class.

Demographic Parity does not condition on any outcome; it equalizes the positive prediction rate.

Predictive Equality conditions on the *true negative* class ($Y = 0$) and equalizes the false positive rate.

This question is a “reverse conditioning trap” because it swaps the conditioning direction relative to Q512.

Question

Demographic parity equalizes:

- A. Precision
- B. Sensitivity
- C. Positive prediction rates
- D. Accuracy

Answer: C

Positive prediction rates

Explanation

Demographic parity (also called statistical parity or independence) requires that the positive prediction rate be equal across groups:

$$P(\hat{Y} = 1 \mid A = 0) = P(\hat{Y} = 1 \mid A = 1)$$

In words: the proportion of individuals predicted positive must be the same for every group, regardless of the true labels.

Why the others are wrong:

Precision is equalized by predictive rate parity.

Sensitivity (TPR) is equalized by equal opportunity.

Accuracy is not the target of any of these standard group-fairness criteria (though it may be equal in some settings).

Demographic parity is the only criterion here that can be evaluated *without* observing ground-truth outcomes, because it depends only on predictions and group membership.

Question

Which pairing is correct?

- A. Demographic Parity – Equal Precision
- B. Equal Opportunity – Equal Sensitivity
- C. Predictive Rate Parity – Equal Recall
- D. Predictive Equality – Equal Precision

Answer: B

Equal Opportunity – Equal Sensitivity

Explanation

The correct pairing is Equal Opportunity – Equal Sensitivity. Equal opportunity requires equal true positive rates (sensitivity/recall) across groups. Why the others are wrong:

- A: Demographic parity equalizes *positive prediction rates*, not precision.
- C: Predictive rate parity equalizes *precision* (positive predictive value), not recall (sensitivity).
- D: Predictive equality equalizes *false positive rates*, not precision.

A useful memory aid:

Criterion	Equalized Metric
Demographic Parity	Positive Prediction Rate
Equal Opportunity	True Positive Rate (Sensitivity/Recall)
Predictive Rate Parity	Precision (PPV)
Predictive Equality	False Positive Rate
Equalized Odds	TPR and FPR

Question

Predictive equality requires equality of:

- A. False positive rates
- B. Precision
- C. Accuracy
- D. Positive prediction rates

Answer: A

False positive rates

Explanation

Predictive equality requires that the false positive rate (FPR) be equal across groups:

$$P(\hat{Y} = 1 \mid Y = 0, A = 0) = P(\hat{Y} = 1 \mid Y = 0, A = 1)$$

In words: among all individuals who are *truly negative*, the proportion who are *incorrectly* predicted positive must be the same for every group. This conditions on the true negative class ($Y = 0$).

Why the others are wrong:

Precision is equalized by predictive rate parity.

Accuracy is not a group-fairness criterion in this family.

Positive prediction rates are equalized by demographic parity.

Predictive equality is sometimes called equal false positive rate and is one half of equalized odds (the other half being equal opportunity).

Question

Equalized odds requires equality of:

- A. Precision and accuracy
- B. Positive prediction rates and precision
- C. True positive rates and false positive rates
- D. Accuracy and sensitivity

Answer: C

True positive rates and false positive rates

Explanation

Equalized odds requires that both the true positive rate (TPR) and the false positive rate (FPR) be equal across groups:

$$P(\hat{Y} = 1 \mid Y = 1, A = 0) = P(\hat{Y} = 1 \mid Y = 1, A = 1)$$

$$P(\hat{Y} = 1 \mid Y = 0, A = 0) = P(\hat{Y} = 1 \mid Y = 0, A = 1)$$

In words: the model must have equal sensitivity *and* equal false positive rate for every group. Equivalently, the prediction $\hat{Y}$ must be independent of the protected attribute A *conditional on the true label* Y .

Why the others are wrong:

- A: Precision and accuracy are not the targets of equalized odds.
- B: Positive prediction rates and precision are targets of demographic parity and predictive rate parity, respectively.
- D: Accuracy and sensitivity are not jointly equalized by equalized odds (only TPR and FPR are).

Equalized odds is a combination of equal opportunity (equal TPR) and predictive equality (equal FPR).

Question

Which fairness criterion can be evaluated without observing ground-truth outcomes?

- A. Equal Opportunity
- B. Predictive Rate Parity
- C. Equalized Odds
- D. Demographic Parity

Answer: D

Demographic Parity

Explanation

Demographic parity can be evaluated without observing ground-truth outcomes because it depends only on the predicted labels and group membership:

$$P(\hat{Y} = 1 \mid A = 0) = P(\hat{Y} = 1 \mid A = 1)$$

No true labels Y are needed.

Why the others require ground truth:

Equal Opportunity requires $Y = 1$ to compute sensitivity.

Predictive Rate Parity requires $Y = 1$ to compute precision.

Equalized Odds requires both $Y = 1$ and $Y = 0$ to compute TPR and FPR.

This is why demographic parity is often attractive in settings where labels are unavailable, delayed, or expensive to obtain. However, it can be satisfied even when the model is inaccurate, which is a known limitation.

Question

A candidate says: "Predictive rate parity equals the share of actual positives correctly identified."
Which metric are they accidentally describing?

- A. Predictive Rate Parity
- B. Demographic Parity
- C. Equal Opportunity
- D. Calibration

Answer: C

Equal Opportunity

Explanation

The phrase "the share of actual positives correctly identified" is the definition of sensitivity (true positive rate / recall):

$$\text{Sensitivity} = P(\hat{Y} = 1 \mid Y = 1)$$

Equal opportunity requires *equal* sensitivity across groups. So the candidate is accidentally describing Equal Opportunity, not predictive rate parity.
What predictive rate parity actually is: Predictive rate parity requires equal precision (positive predictive value) across groups:

$$P(Y = 1 \mid \hat{Y} = 1, A = 0) = P(Y = 1 \mid \hat{Y} = 1, A = 1)$$

Precision is "the share of *predicted* positives that are actually positive," which is the reverse conditioning of sensitivity.
Why the others are wrong:

- A: Predictive rate parity is precision, not sensitivity.
- B: Demographic parity is the positive prediction rate, $P(\hat{Y} = 1 \mid A)$.
- D: Calibration refers to the agreement between predicted probabilities and observed frequencies, not this quantity.

Key distinction:

Watch the video